

The Mathematics and Physics of Reusable Launch Vehicles: From the Rocket Equation to Propulsive Landing

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Abstract

Reusable launch vehicles have transformed the economics of spaceflight in little more than a decade. This paper presents a self-contained mathematical treatment of the physics that governs launch, staging, atmospheric re-entry, and propulsive recovery of orbital-class boosters, and anchors every result in real vehicles: SpaceX’s Falcon 9 and Starship/Super Heavy, Blue Origin’s New Shepard and New Glenn, and Rocket Lab’s Electron and Neutron. We begin with the Tsiolkovsky rocket equation and the loss budget for reaching low Earth orbit, derive the optimal staging problem and quantify the payload penalty incurred by reserving propellant for recovery, and then treat the recovery flight itself: boostback and entry burns, hypersonic aerodynamic deceleration, grid-fin control authority, convective heating, and the “hoverslam” landing burn as an optimal-control problem solved in real time by lossless convexification. We close with scaling laws that explain why reusability is easier for large vehicles, structural and thermal fatigue considerations for multi-flight life, and a cost model showing when reuse pays. Worked numerical examples reproduce the published performance of Falcon 9 to within a few percent using only first-principles physics.

1 Introduction

For the first sixty years of orbital spaceflight, nearly every launch vehicle was discarded after a single use. A Saturn V, a Delta IV, or an Ariane 5 was, in effect, a jet airliner that was scrapped after one flight. The obvious economic absurdity of this practice was well understood; what was not understood was how to close the engineering problem of bringing back a booster that had just accelerated to several times the speed of sound and was hundreds of kilometres downrange.

The problem is fundamentally one of *mass and energy*. A rocket stage is dominated by propellant (typically 90 % to 95 % of its liftoff mass), and the equations governing its motion are exponential in the ratio of initial to final mass. Every kilogram of hardware or propellant reserved for recovery is a kilogram not delivered to orbit, and because of the exponential, the penalty compounds. Conversely, every kilogram of orbit-bound structure that is *not* thrown away is capital that can be amortised over many flights.

The first fully successful landing of an orbital-class booster occurred on 21 December 2015, when SpaceX’s Falcon 9 first stage returned to Landing Zone 1 at Cape Canaveral. The first re-flight of a recovered booster followed in March 2017. By mid-2026 individual Falcon 9 boosters had exceeded thirty flights each, the Falcon family had performed well over five hundred booster landings, SpaceX had caught the 70m Super Heavy booster in mid-air with its launch tower, and Blue Origin had recovered a New Glenn first stage on a drone ship. Reusability had gone from aspiration to the default.

This paper aims to give a reader with an undergraduate background in physics and calculus a complete, quantitative picture of *why* reusable rockets work, *how much* they cost in performance, and *what mathematics* is executed on the flight computer during the final seconds before

touchdown. We deliberately keep the treatment analytic where possible, resorting to numerical examples only to connect the formulas to actual flight data.

1.1 Notation and conventions

Throughout, $g_0 = 9.806\,65\,\text{m/s}^2$ is standard gravity, used to convert specific impulse (in seconds) to effective exhaust velocity; $\mu_\oplus = 3.986 \times 10^{14}\,\text{m}^3/\text{s}^2$ is Earth’s gravitational parameter; $R_\oplus = 6371\,\text{km}$ is the mean Earth radius. Vectors are set in **bold**. Masses are in kilograms unless stated. The word *booster* refers to the first stage of a multistage vehicle. Vehicle data are drawn from public manufacturer specifications and are approximate; where different sources disagree we quote representative values and indicate the range.

Table 1: Reference data for vehicles discussed in this paper. Values are representative public figures and vary by block, mission, and source.

Vehicle	Liftoff mass [t]	S1 thrust (SL) [kN]	S1 I_{sp} SL/vac [s]	Engines (S1)	Recovery mode
Falcon 9 Block 5	549	7 600	283 / 312	9 Merlin 1D	Legs, RTLS/ASDS
Falcon Heavy	1 420	22 800	283 / 312	27 Merlin 1D	Legs (3 cores)
Starship / Super Heavy	~5 000	~74 000*	327 / 350	33 Raptor	Tower catch
New Shepard	~75	490	~310 (vac)	1 BE-3	Legs, suborbital
New Glenn	~1 500	17 100	~310 / 330	7 BE-4	Legs, ship
Electron	13	190	303 (vac)	9 Rutherford	Parachute/splash
Neutron	~480	7 300	~300 / 320	9 Archimedes	Legs, RTLS/ship

*Super Heavy liftoff thrust $\approx 7500\,\text{tf} \approx 74\,\text{MN}$ with Raptor 2/3; higher for later blocks.

2 The Rocket Equation and the Cost of Reaching Orbit

2.1 Derivation of the Tsiolkovsky equation

Consider a rocket of instantaneous mass $m(t)$ expelling propellant at mass flow rate $\dot{m}_p = -\dot{m} > 0$ with exhaust velocity v_e relative to the vehicle. In the absence of external forces, conservation of momentum over an interval dt gives

$$m\,dv = -v_e\,dm, \quad (1)$$

so that integrating from initial mass m_0 to final mass m_f ,

$$\boxed{\Delta v = v_e \ln \frac{m_0}{m_f} = g_0 I_{\text{sp}} \ln \frac{m_0}{m_f}} \quad (2)$$

Equation (2) is the Tsiolkovsky rocket equation. The quantity $v_e = g_0 I_{\text{sp}}$ is the *effective* exhaust velocity, which accounts for pressure thrust at the nozzle exit. Inverting,

$$\frac{m_0}{m_f} = \exp\left(\frac{\Delta v}{v_e}\right), \quad \frac{m_p}{m_0} = 1 - \exp\left(-\frac{\Delta v}{v_e}\right), \quad (3)$$

where $m_p = m_0 - m_f$ is the propellant consumed. The exponential is the entire story of launch-vehicle design: because $\Delta v_{\text{LEO}} \approx 9.4\,\text{km/s}$ while chemical v_e is at most $\sim 4.5\,\text{km/s}$, a single-stage vehicle would require $m_0/m_f \approx e^{2.1} \approx 8$, i.e. 88% propellant by mass before any payload is carried.

2.2 Specific impulse of real engines

Specific impulse depends on the propellant combination, chamber pressure, and nozzle expansion ratio. Sea-level values are lower than vacuum values because ambient pressure acts on the nozzle exit area:

$$F = \dot{m}_p u_e + (p_e - p_a)A_e, \quad I_{\text{sp}} = \frac{F}{\dot{m}_p g_0}. \quad (4)$$

Table 2: Representative performance of engines used on reusable boosters.

Engine	Propellants	Cycle	Thrust SL [kN]	I_{sp} SL [s]	I_{sp} vac [s]
Merlin 1D (Falcon 9)	RP-1 / LOX	Gas generator	845	283	312
Merlin 1D Vacuum	RP-1 / LOX	Gas generator	n/a	n/a	348
Raptor 2 (Starship)	CH ₄ / LOX	Full-flow staged	2 300	327	350
Raptor 3	CH ₄ / LOX	Full-flow staged	~2 750	~330	~355
BE-4 (New Glenn)	CH ₄ / LOX	Ox-rich staged	2 400	~310	~335
BE-3PM (New Shepard)	LH ₂ / LOX	Tap-off	490	n/a	~310
Rutherford (Electron)	RP-1 / LOX	Electric pump	25	303	343
Archimedes (Neutron)	CH ₄ / LOX	Ox-rich staged	730	~300	~320

Two observations matter for reusability. First, methane (CH₄) has become the propellant of choice for new reusable engines despite RP-1's higher density, because methane burns cleaner (no coking of injectors and turbine blades), can be stored at temperatures close to LOX, and is producible on Mars. Second, high-chamber-pressure staged-combustion cycles (Raptor at ~300 bar, BE-4 at ~135 bar) deliver high thrust density, which is essential when a landing booster must carry only a fraction of its engines lit.

2.3 The Δv budget to low Earth orbit

The idealised orbital velocity at altitude h is

$$v_{\text{circ}} = \sqrt{\frac{\mu_{\oplus}}{R_{\oplus} + h}} \approx 7.79 \text{ km/s} \quad (h = 200 \text{ km}). \quad (5)$$

The actual Δv a vehicle must generate is larger because of losses integrated along the trajectory. Writing the equation of motion along the velocity vector,

$$\frac{dv}{dt} = \frac{F \cos \alpha}{m} - g \sin \gamma - \frac{D}{m}, \quad (6)$$

where γ is the flight-path angle, α the angle of attack, and D the drag, integrating gives the budget

$$\Delta v_{\text{ideal}} = \underbrace{\int \frac{F}{m} dt}_{\text{delivered}} = v_{\text{orbit}} + \underbrace{\int g \sin \gamma dt}_{\Delta v_g} + \underbrace{\int \frac{D}{m} dt}_{\Delta v_D} + \underbrace{\int \frac{F(1 - \cos \alpha)}{m} dt}_{\Delta v_{\text{steer}}} - \Delta v_{\text{rot}}, \quad (7)$$

with $\Delta v_{\text{rot}} = \omega_{\oplus} R_{\oplus} \cos \phi \approx 0.41 \text{ km/s}$ for an eastward launch from latitude $\phi = 28.5^\circ$ (Cape Canaveral).

Table 3: Typical loss budget for a Falcon-9-class vehicle to a 200 km circular orbit.

Term	Δv [km/s]
Orbital velocity	7.79
Gravity loss Δv_g	1.20 – 1.50
Drag loss Δv_D	0.05 – 0.15
Steering loss	0.10 – 0.25
Earth rotation credit	−0.41
Total required	9.0 – 9.4

Gravity loss dominates. It is proportional to the time spent with a vertical velocity component and is therefore reduced by high initial thrust-to-weight ratio T/W and by pitching over early. This is the first place where reusability trades against performance: a recoverable booster must stage *earlier* (lower velocity, so it can fly back or survive re-entry), which pushes more of the gravity loss onto the upper stage.

2.4 Gravity loss in closed form for a vertical ascent

For a purely vertical burn at constant thrust F and constant $\dot{m}_p$, with $m(t) = m_0 - \dot{m}_p t$, the velocity at burnout t_b is

$$v(t_b) = v_e \ln \frac{m_0}{m_f} - g_0 t_b, \quad t_b = \frac{m_p}{\dot{m}_p} = \frac{v_e m_p}{F}. \quad (8)$$

Writing $n_0 = F/(m_0 g_0)$ for the initial thrust-to-weight ratio and $\zeta = m_p/m_0$ for the propellant fraction,

$$\frac{\Delta v_g}{v_e} = \frac{\zeta}{n_0}. \quad (9)$$

For Falcon 9 at liftoff, $n_0 \approx 7.6 \text{ MN}/(549 \text{ t} \cdot g_0) \approx 1.41$. If the first stage burned vertically to depletion with $\zeta \approx 0.72$ (booster propellant over total liftoff mass), Eq. (9) would give $\Delta v_g \approx 0.51 v_e \approx 1.5 \text{ km/s}$, an overestimate since real trajectories pitch over, but the correct order of magnitude and the correct scaling: gravity loss falls as $1/n_0$.

Example 2.1 (Falcon 9 Merlin throttle-up). *Merlin 1D thrust rose from 756 kN (v1.1) to 845 kN (Block 5). At fixed liftoff mass this raises n_0 from 1.26 to 1.41, cutting the vertical-ascent gravity loss by about 10%, i.e. roughly 150 m/s. Via Eq. (3) on the second stage, 150 m/s corresponds to about 4% of second-stage propellant, or roughly 700 kg of additional payload to LEO. Engine thrust upgrades were one of the main levers SpaceX used to buy back the payload cost of reusability.*

3 Multistage Vehicles and the Reusability Penalty

3.1 Structural coefficient and the payload fraction

Define for stage i the structural (dry) mass $m_{s,i}$, propellant mass $m_{p,i}$, and the *structural coefficient*

$$\varepsilon_i = \frac{m_{s,i}}{m_{s,i} + m_{p,i}}. \quad (10)$$

Modern kerosene boosters achieve $\varepsilon \approx 0.05$ – 0.06 ; Falcon 9’s first stage, with $m_s \approx 25.6 \text{ t}$ and $m_p \approx 395 \text{ t}$, has $\varepsilon_1 \approx 0.061$ *including* landing legs, grid fins, and cold-gas thrusters. Its second stage ($m_s \approx 4 \text{ t}$, $m_p \approx 92 \text{ t}$) has $\varepsilon_2 \approx 0.042$.

For an N -stage vehicle carrying payload m_L , with $m_{0,i}$ the total mass at ignition of stage i , the rocket equation applies stage by stage:

$$\Delta v_{\text{total}} = \sum_{i=1}^N v_{e,i} \ln \frac{m_{0,i}}{m_{0,i} - m_{p,i}}. \quad (11)$$

Define the *payload ratio* of stage i as $\lambda_i = m_{0,i+1}/m_{0,i}$ (with $m_{0,N+1} = m_L$). Then $m_{0,i}/(m_{0,i} - m_{p,i}) = 1/(\varepsilon_i + \lambda_i - \varepsilon_i \lambda_i)$ and the overall payload fraction is $m_L/m_{0,1} = \prod_i \lambda_i$.

3.2 Optimal staging

For similar stages (v_e and ε equal across stages) the Lagrangian maximisation of $\prod \lambda_i$ subject to Eq. (11) yields equal Δv per stage. For dissimilar stages, maximising $\prod_i \lambda_i$ with a Lagrange multiplier η on the Δv constraint gives, for each stage, an optimal mass ratio $n_i = m_{0,i}/(m_{0,i} - m_{p,i})$ of

$$n_i = \frac{v_{e,i} \eta - 1}{v_{e,i} \varepsilon_i \eta}, \quad \text{with } \eta \text{ fixed by } \sum_{i=1}^N v_{e,i} \ln n_i = \Delta v_{\text{total}}, \quad (12)$$

which, given the higher I_{sp} of upper stages, pushes a larger share of Δv onto the upper stage. A fully expendable Falcon 9 stages at about 3.1 km/s inertial; the reusable profile stages at 2.0 km/s to 2.4 km/s (drone-ship landing) or 1.6 km/s to 1.8 km/s (return to launch site), far from the expendable optimum. The vehicle is deliberately *de-optimised* for performance so that the booster's energy at separation is low enough to be dissipated safely.

3.3 The reusability penalty

Let the booster reserve a Δv_{res} for recovery manoeuvres. The propellant required is

$$m_{p,\text{res}} = m_{\text{land}} \left[\exp\left(\frac{\Delta v_{\text{res}}}{v_{e,1}}\right) - 1 \right], \quad (13)$$

where m_{land} is the booster mass at touchdown ($\approx m_{s,1}$ plus residuals). This propellant is carried through the ascent burn but not used for it, so the effective booster propellant for ascent is $m_{p,1} - m_{p,\text{res}}$.

Example 3.1 (Falcon 9 drone-ship recovery). *Take $m_{\text{land}} \approx 27$ t (dry mass plus 1.5 t residuals), $v_{e,1} = 2.95$ km/s (mean of SL and vacuum I_{sp} over the recovery burns). A typical ASDS profile uses an entry burn of ~ 600 m/s and a landing burn of ~ 300 m/s (plus gravity loss during the landing burn), so $\Delta v_{\text{res}} \approx 1.0$ km/s:*

$$m_{p,\text{res}} \approx 27 [e^{1.0/2.95} - 1] \approx 11 \text{ t}.$$

For return to launch site (RTLS), a boostback burn of ~ 800 m/s is added, giving $\Delta v_{\text{res}} \approx 1.8$ km/s and $m_{p,\text{res}} \approx 27 [e^{1.8/2.95} - 1] \approx 22.5$ t.

The effect on payload is amplified through the second stage. To first order, a booster Δv deficit δ must be made up by the upper stage, whose payload sensitivity is

$$\frac{\partial m_L}{\partial \Delta v_2} \approx -\frac{m_{0,2}}{v_{e,2}} e^{-\Delta v_2/v_{e,2}}, \quad (14)$$

so a 600 m/s reduction in booster contribution costs on the order of $110 \text{ t} \times 0.6/3.41 \times e^{-6.3/3.41} \approx 3 \text{ t}$ of payload at the LEO reference point. This matches the published numbers: Falcon 9 lifts 22.8 t to LEO expendable, roughly 17 t with drone-ship recovery (a 25 % to 30 % penalty), and roughly 12 t to 13 t for RTLS (a ~ 40 % penalty). The penalty is smaller in percentage terms for

high-energy orbits, where the second stage’s exponential sensitivity means the mission is already payload-light.

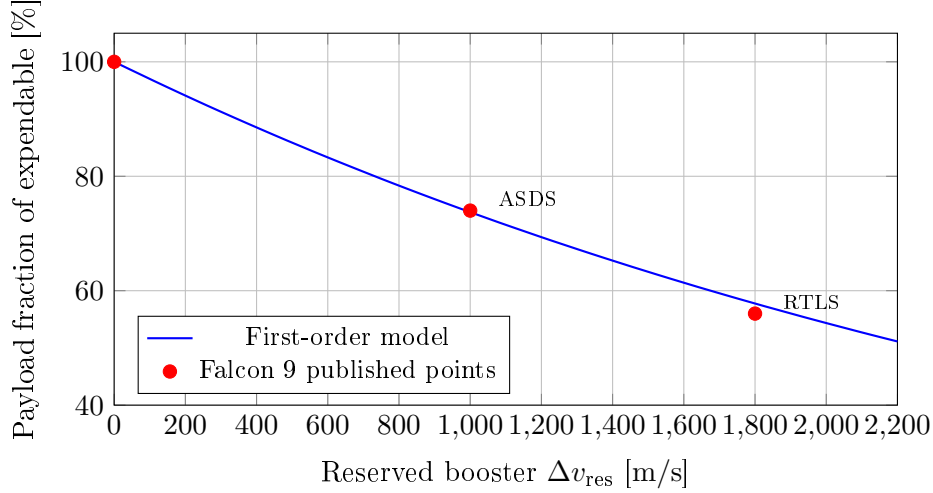


Figure 1: Payload to LEO relative to expendable configuration as a function of the Δv reserved in the booster for recovery. The exponential model $m_L/m_{L,\text{exp}} = \exp(-\Delta v_{\text{res}}/v_{e,\text{eff}})$ with $v_{e,\text{eff}} \approx 3.3 \text{ km/s}$ reproduces the SpaceX-quoted performance points.

3.4 Why the second stage is (usually) not recovered

The same arithmetic explains why Falcon 9’s upper stage is expended. It reaches orbital velocity, so recovery requires surviving a full orbital re-entry (energy $\frac{1}{2}v^2 \approx 30 \text{ MJ/kg}$, versus $\sim 2 \text{ MJ/kg}$ for the booster) and carrying a thermal-protection system, aerodynamic control surfaces, and landing propellant, all of which come *directly* off the payload with a one-to-one exchange ratio. For a stage with $\varepsilon_2 = 0.04$, adding 3 t of recovery hardware to a 4 t dry mass nearly doubles its structural coefficient and removes 3 t from a 17 t payload. Starship attacks this problem by making the upper stage so large (1300 t propellant, $\sim 100 \text{ t}+$ dry) that a heat shield and flaps are a modest fraction of its mass, and by giving up expendable-mode performance entirely in exchange for full reusability.

4 Ascent Trajectory and Staging Conditions

4.1 Equations of motion in a rotating, spherical Earth frame

In a planar model with radial distance r , flight-path angle γ (measured from the local horizontal), and speed v , the point-mass equations are

$$\dot{v} = \frac{F \cos \alpha - D}{m} - g(r) \sin \gamma, \quad (15)$$

$$v\dot{\gamma} = \frac{F \sin \alpha + L}{m} - \left(g(r) - \frac{v^2}{r} \right) \cos \gamma, \quad (16)$$

$$\dot{r} = v \sin \gamma, \quad \dot{x} = \frac{R_{\oplus}}{r} v \cos \gamma, \quad \dot{m} = -\frac{F}{v_e}, \quad (17)$$

where $g(r) = \mu_{\oplus}/r^2$, $D = \frac{1}{2}\rho v^2 C_D A$, $L = \frac{1}{2}\rho v^2 C_L A$, and x is downrange distance. The term v^2/r is the centrifugal relief that eventually balances gravity at orbital speed.

4.2 The gravity turn

With $\alpha = 0$ and $L = 0$, the thrust vector remains aligned with velocity and Eq. (16) reduces to

$$\dot{\gamma} = -\left(\frac{g}{v} - \frac{v}{r}\right) \cos \gamma. \quad (18)$$

This is the *gravity turn*: after a small initial pitch kick, gravity alone rotates the velocity vector toward horizontal, with zero steering loss. Falcon 9 flies a near-gravity-turn for most of first-stage flight; the pitch kick occurs at about $T+10$ s and the vehicle passes through max- Q (maximum dynamic pressure, $q = \frac{1}{2}\rho v^2 \approx 35$ kPa) at roughly $T+70$ s and 12 km altitude, where the engines are briefly throttled to limit structural loads.

4.3 Atmosphere model

For analytic work we use the exponential atmosphere

$$\rho(h) = \rho_0 e^{-h/H}, \quad \rho_0 = 1.225 \text{ kg/m}^3, \quad H \approx 7.2 \text{ km} \quad (19)$$

(scale height varies from about 8.5 km at sea level to 6 km in the stratosphere; 7.2 km is a good average to 80 km).

4.4 Staging state for a reusable booster

The choice of main-engine cutoff (MECO) state fixes everything about the recovery. Table 4 lists representative MECO conditions.

Table 4: Representative first-stage cutoff conditions (values reconstructed from public webcast telemetry; ranges reflect mission variation).

Profile	MECO time [s]	Altitude [km]	Speed [km/s]	Landing downrange [km]
Falcon 9 RTLS	135–145	60–70	1.6–1.9	0 (LZ-1 / LZ-4)
Falcon 9 ASDS (LEO)	150–160	70–80	2.2–2.5	500–650
Falcon 9 ASDS (GTO)	155–165	75–85	2.5–2.8	600–700
Falcon 9 expendable	170–180	90–100	3.0–3.4	(n/a)
Super Heavy (tower catch)	~150	~65	~1.6–1.9	0 (launch site)
New Glenn (ship)	~190	~95	~3.0	~600

The booster’s specific mechanical energy at MECO,

$$\mathcal{E} = \frac{v^2}{2} + g_0 h \quad (\text{flat-Earth approximation}), \quad (20)$$

is 3.3 MJ/kg for a typical ASDS profile and 2.2 MJ/kg for RTLS. Of this, only a small fraction can be removed propulsively; the rest must be dissipated by the atmosphere. For comparison, orbital specific energy is ~ 32 MJ/kg, an order of magnitude larger.

4.5 Ballistic apogee and the “coast” phase

Neglecting drag (valid above 70 km) and treating gravity as uniform over the short arc, the booster after MECO with vertical velocity v_z and horizontal v_x reaches apogee after $t_a = v_z/g_0$ at height $h_{\max} = h_{\text{MECO}} + v_z^2/(2g_0)$. For an ASDS profile with $v = 2.3$ km/s at $\gamma = 45^\circ$ from $h = 75$ km: $v_z \approx 1.63$ km/s, $h_{\max} \approx 75 + 135 = 210$ km, $t_a \approx 165$ s. Falcon 9 boosters routinely reach 120 km to 200 km apogee on drone-ship missions, consistent with this estimate once the lower actual γ is accounted for. The total flight time from liftoff to landing is about 8.5 min to 9.5 min for ASDS and 7.5 min to 8 min for RTLS.

5 The Recovery Flight: Boostback, Entry, and Aerodynamic Deceleration

Immediately after stage separation, a Falcon 9 booster performs the sequence in Table 5. Each phase is governed by a different piece of physics, and we treat them in turn.

Table 5: Falcon 9 booster recovery sequence (ASDS profile; RTLS adds a boostback burn).

Phase	Time after liftoff	Engines	Physics
Stage separation	$\sim T+155$ s	n/a	Pneumatic pushers; nitrogen cold-gas thrusters flip booster $\sim 180^\circ$
Boostback burn (RTLS only)	T+165 to T+215 s	3	Reverses horizontal velocity; $\Delta v \approx 600\text{--}900$ m/s
Apogee, grid-fin deploy	T+210 to T+260 s	n/a	Exoatmospheric coast; fins unfold in vacuum
Entry burn	T+390 to T+415 s	3	Reduces speed from ~ 1.7 to ~ 1.0 km/s before dense air; $\Delta v \approx 500\text{--}700$ m/s
Aerodynamic descent	T+415 to T+500 s	n/a	Body-lift and grid-fin guidance; terminal $v \approx 250\text{--}300$ m/s
Landing burn	T+500 to T+525 s	1 (or 3 $\rightarrow$ 1)	“Hoverslam”; $\Delta v \approx 300\text{--}400$ m/s incl. gravity loss
Touchdown	$\sim T+525$ s	n/a	$v < 2$ m/s, legs absorb residual energy

5.1 The flip manoeuvre

After separation the booster must rotate engines-forward. With moment of inertia $I \approx \frac{1}{12}m\ell^2$ for a slender body ($m = 30$ t propellant-loaded, $\ell = 41$ m gives $I \approx 4.2 \times 10^6$ kg m²), and cold-gas thrusters delivering torque $\tau = F_{\text{thr}}d$ with $F_{\text{thr}} \sim 1$ kN per thruster at moment arm $d \approx 15$ m (two thrusters firing), the angular acceleration is $\ddot{\theta} = \tau/I \approx 7 \times 10^{-3}$ rad/s². A bang-bang rotation through π radians (accelerate half-way, decelerate half-way) takes

$$t_{\text{flip}} = 2\sqrt{\frac{\pi}{\ddot{\theta}}} \approx 42 \text{ s}, \quad (21)$$

which is consistent with the observed ~ 30 s to 40 s between separation and boostback ignition. Super Heavy instead performs the flip using its main engines’ thrust-vector control during “hot staging”, turning the turnaround into a powered manoeuvre lasting only a few seconds.

5.2 Boostback burn geometry

For RTLS the booster must cancel its downrange velocity v_x and add a return velocity $v_{x,\text{ret}}$ sufficient to fly back over the ballistic arc. Treating the return as a ballistic trajectory from apogee height with total time-of-flight t_f and required range x , the horizontal velocity needed after boostback is

$$v_{x,\text{ret}} = \frac{x_{\text{MECO}} + v_x t_{\text{bb}}}{t_f - t_{\text{bb}}}, \quad (22)$$

where t_{bb} is the time to boostback and the numerator accounts for the additional downrange drift before the burn. For a typical RTLS ($x_{\text{MECO}} \approx 40$ km, $v_x \approx 1.2$ km/s, $t_{\text{bb}} = 30$ s, $t_f \approx 380$ s), $v_{x,\text{ret}} \approx 220$ m/s and the boostback $\Delta v \approx v_x + v_{x,\text{ret}} \approx 1.4$ km/s in the flat-Earth approximation; real profiles have lower v_x at MECO (the RTLS trajectory is lofted, with γ closer to $60^\circ\text{--}70^\circ$), bringing the burn down to 600 m/s to 900 m/s. Note the burn does not simply reverse velocity: it is aimed to place the booster’s ballistic impact point on the landing pad, with the entry and landing burns then providing the remaining correction.

5.3 Re-entry dynamics: the Allen–Eggers solution

The central problem of booster recovery is that the vehicle re-enters engines-first at 1.5 km/s to 2.5 km/s into the sensible atmosphere. Consider a ballistic entry at constant flight-path angle γ_e (measured below horizontal) with drag as the only significant force. Then

$$m \frac{dv}{dt} = -\frac{1}{2} \rho v^2 C_D A, \quad \frac{dh}{dt} = -v \sin \gamma_e. \quad (23)$$

Dividing, $dv/dh = \rho v C_D A / (2m \sin \gamma_e)$. Substituting Eq. (19) and integrating from $h = \infty$ (where $v = v_e$),

$$v(h) = v_e \exp \left[-\frac{\rho_0 H}{2\beta \sin \gamma_e} e^{-h/H} \right], \quad \beta \equiv \frac{m}{C_D A} \quad (24)$$

where β is the *ballistic coefficient*. Differentiating the deceleration $a = \rho v^2 / (2\beta)$ with respect to h and setting it to zero gives the altitude and magnitude of peak deceleration:

$$h_{\text{peak}} = H \ln \left(\frac{\rho_0 H}{\beta \sin \gamma_e} \right), \quad a_{\text{max}} = \frac{v_e^2 \sin \gamma_e}{2eH}. \quad (25)$$

Remarkably, a_{max} does not depend on β : a heavier or more streamlined body simply decelerates *lower* in the atmosphere, not less.

Example 5.1 (Falcon 9 without an entry burn). For $v_e = 2.3$ km/s, $\gamma_e = 60^\circ$: $a_{\text{max}} = (2300^2 \times 0.866) / (2 \times 2.718 \times 7200) \approx 117 \text{ m/s}^2 \approx 12g$. The booster ($m = 30 \text{ t}$, $A = \pi(1.83)^2 = 10.5 \text{ m}^2$, $C_D \approx 1.0$ engines-first with a turbulent base) has $\beta \approx 2900 \text{ kg/m}^2$, giving $h_{\text{peak}} = 7.2 \ln(1.225 \times 7200 / (2900 \times 0.866)) \approx 9 \text{ km}$. The dynamic pressure at peak would be $q = \beta a_{\text{max}} \approx 340 \text{ kPa}$, roughly ten times the ascent max- Q and well beyond the structure’s capability. An aluminium-lithium tank designed to 35 kPa to 45 kPa on ascent cannot survive this.

5.4 The entry burn as a loads-management tool

The entry burn exists to reduce v_e before Eq. (25) takes over. Since $a_{\text{max}} \propto v_e^2$, cutting the entry speed from 2.3 to 1.2 km/s reduces peak deceleration by a factor of 3.7, to $\sim 3.5g$ and peak q to $\sim 90 \text{ kPa}$. The burn also raises the effective ballistic altitude at which deceleration begins: a three-engine burn firing into the oncoming hypersonic flow creates a large low-density plume ahead of the vehicle that effectively increases the drag area during the burn and shields the engines from stagnation heating. SpaceX telemetry shows the entry burn beginning at about 70 km and ending near 40 km, exactly where Eq. (19) gives $\rho \approx 4 \times 10^{-3} \text{ kg/m}^3$ and drag first becomes significant.

The propellant cost is the trade. Using $m_{\text{land}} = 27 \text{ t}$ and $v_e = 2.9 \text{ km/s}$, a 600 m/s entry burn requires $\approx 6.3 \text{ t}$ of propellant, about 1.6 % of the booster’s load, and the largest single item in the recovery budget for ASDS missions. Super Heavy, which stages at lower velocity and has a lower ballistic coefficient (larger diameter, $\beta \approx 1900 \text{ kg/m}^2$ with $m \approx 200 \text{ t}$ at entry), does not use an entry burn at all; it accepts a higher peak heating load on its engine bay and stainless-steel tanks.

5.5 Aerodynamic heating

The convective stagnation-point heat flux is well approximated by the Sutton–Graves relation

$$\dot{q}_s = k \sqrt{\frac{\rho}{r_n}} v^3, \quad k = 1.74 \times 10^{-4} \text{ kg}^{1/2}/\text{m} \text{ (air, SI)}, \quad (26)$$

with r_n the effective nose radius. For the booster the relevant surfaces are the engine bells and the octaweb base, with an effective $r_n \sim 1 \text{ m}$ for the blunt base. Substituting the Allen–Eggers

velocity profile and maximising, peak heating occurs at $v = v_e/e^{1/6} \approx 0.85v_e$, at an altitude $H \ln(3\rho_0 H/(\beta \sin \gamma_e))$, i.e. about one scale height above peak deceleration. With the entry burn ($v_e \approx 1.2 \text{ km/s}$) and $\rho \approx 0.02 \text{ kg/m}^3$ at 30 km,

$$\dot{q}_s \approx 1.74 \times 10^{-4} \sqrt{0.02/1} \times (1000)^3 \approx 25 \text{ kW/m}^2,$$

modest compared with an orbital capsule (1 MW/m^2 to 5 MW/m^2) and manageable by the thermal mass of the engine hardware plus ablative cork and later titanium and steel shielding on the octaweb. The integrated heat load $\int \dot{q} dt \sim 1 \text{ MJ/m}^2$ is why a Falcon 9 booster can be re-flown with only inspection rather than refurbishment of its base.

Grid fins are the exception: mounted at the top of the vehicle where the flow is attached and fully stagnated on their leading edges ($r_n \sim 1 \text{ cm}$), they see $\dot{q}_s$ an order of magnitude higher. The original aluminium fins with ablative coating were damaged on high-energy (GTO) entries and were replaced in 2017 by single-piece cast-and-cut titanium fins, which tolerate the temperatures without coating and are reused indefinitely.

5.6 Grid fin aerodynamics and control authority

A grid fin is a lattice of thin planes forming many small cells. Its principal advantages for a landing booster are (i) a very short chord, so the centre of pressure barely moves with Mach number and the hinge moment stays small, allowing compact actuators; (ii) high lift-curve slope at supersonic speed; and (iii) folding flush against the body during ascent. Its main disadvantage is a transonic drag peak when cell flow chokes ($M \approx 0.8\text{--}1.2$), during which the fin's normal-force effectiveness drops sharply.

For a grid fin of frontal area S_f , the normal force per unit deflection δ is

$$N = \frac{1}{2} \rho v^2 S_f C_{N\delta} \delta, \quad (27)$$

with $C_{N\delta}$ of order 2–3 per radian at supersonic Mach in the linear range. Four fins at moment arm $\ell_f \approx 20 \text{ m}$ from the centre of mass (which is low, near the engines) provide a pitching moment $M = 2N\ell_f$ for a pitch command (two fins act in pitch, two in yaw; all four in roll). At 20 km altitude, $\rho = 0.09 \text{ kg/m}^3$, $v = 500 \text{ m/s}$, $S_f = 1.5 \text{ m}^2$, $\delta = 10^\circ$: $N \approx 0.5 \times 0.09 \times 500^2 \times 1.5 \times 2.5 \times 0.175 \approx 7.4 \text{ kN}$ per fin and $M \approx 300 \text{ kN m}$. Against the entry-phase moment of inertia ($I \approx 4 \times 10^6 \text{ kg m}^2$) this gives $\ddot{\theta} \approx 0.075 \text{ rad/s}^2$, enough to command several degrees of angle of attack in a fraction of a second, which is the mechanism by which the booster steers.

5.7 Body lift and the entry corridor

The booster is flown at a small angle of attack during atmospheric descent so that body lift $L = \frac{1}{2} \rho v^2 C_{L\alpha} \alpha A_{\text{ref}}$ produces a horizontal acceleration that can be steered. A cylinder of fineness ratio $\ell/d \approx 11$ has $C_{L\alpha} \approx 2$ per radian at small α (slender-body theory gives $C_{N\alpha} = 2$ exactly in the slender limit), so at $\alpha = 5^\circ$, $q = 20 \text{ kPa}$: $L \approx 20\,000 \times 2 \times 0.087 \times 10.5 \approx 37 \text{ kN}$, or a lateral acceleration of $\sim 1.2 \text{ m/s}^2$ on a 30 t vehicle. Integrated over the 80 s atmospheric descent this yields several kilometres of cross-range, which is why the booster's ballistic impact point is deliberately placed *short* of the drone ship: it glides the last few kilometres, and a loss of control causes it to fall into the sea rather than onto the ship.

6 The Landing Burn: Powered Descent as an Optimal-Control Problem

6.1 Why a rocket cannot hover

A Merlin 1D throttles to a minimum of about 40% of rated thrust, i.e. $\sim 360 \text{ kN}$ at sea level. A landing booster with $m \approx 27 \text{ t}$ weighs $\sim 265 \text{ kN}$. Even one engine at minimum throttle has

thrust-to-weight $T/W \approx 1.36 > 1$: the vehicle cannot hover and will accelerate upward if the engine is left running. Once the landing burn starts, the only way to arrive at the ground with zero velocity is to time ignition such that velocity and altitude reach zero simultaneously. This is the “hoverslam” (SpaceX’s term) or “suicide burn.” Super Heavy (33 Raptors, but only 3–13 lit for landing, throttleable to $\sim 40\%$) and New Shepard (single BE-3 throttling to $\sim 20\%$) can hover; New Shepard uses this margin to descend slowly and precisely, while Super Heavy uses it to null out the final position errors required for a tower catch.

6.2 One-dimensional constant-thrust solution

Let the booster be at altitude h_0 descending at speed v_0 (positive downward), with constant net upward acceleration $a_{\text{net}} = T/m - g$ during the burn (treating m as constant for the short burn). Kinematics require $v_0^2 = 2a_{\text{net}}h_0$, so ignition must occur at

$$h_{\text{ign}} = \frac{v_0^2}{2a_{\text{net}}}, \quad t_{\text{burn}} = \frac{v_0}{a_{\text{net}}}. \quad (28)$$

For Falcon 9 with a single engine at $\sim 70\%$: $T = 590 \text{ kN}$, $a_{\text{net}} = 590/27 - 9.81 \approx 12 \text{ m/s}^2$; with $v_0 = 300 \text{ m/s}$ (terminal velocity after aerodynamic descent, including the residual atmospheric drag as part of the balance),

$$h_{\text{ign}} \approx \frac{300^2}{2 \times 12} \approx 3.7 \text{ km}, \quad t_{\text{burn}} \approx 25 \text{ s}.$$

This matches the observed 20 s to 30 s landing burns starting at 3 km to 5 km altitude, and is why the three-engine landing burn (used on heavy GTO missions) is so short: with $a_{\text{net}} \approx 50 \text{ m/s}^2$ the burn lasts only $\sim 8 \text{ s}$ and ignites at 1 km, with a hand-off to one engine for the final seconds.

The gravity loss during the burn is $g_0 t_{\text{burn}} \approx 250 \text{ m/s}$ for the single-engine case versus $\sim 80 \text{ m/s}$ for three engines; the three-engine burn saves roughly 1.5 t of propellant at the cost of a harder, less forgiving timing problem. This is a direct instance of Eq. (9): high T/W minimises gravity loss.

6.3 Sensitivity: why timing must be exact

Perturbing the ignition altitude by δh at fixed v_0 changes the touchdown velocity. From $v_{\text{td}}^2 = v_0^2 - 2a_{\text{net}}(h_{\text{ign}} + \delta h)$, expanding near $v_{\text{td}} = 0$:

$$v_{\text{td}} \approx \sqrt{2a_{\text{net}}|\delta h|}. \quad (29)$$

Igniting 20 m late with $a_{\text{net}} = 12 \text{ m/s}^2$ produces a touchdown at 22 m/s, which is catastrophic. Igniting early produces a hover point above the pad with no propellant to descend from. Throttle authority is what makes the problem solvable: the engine runs at $\sim 70\%$ nominally so that it can throttle up to 100% or down to 40%, giving $a_{\text{net}} \in [3.5, 21] \text{ m/s}^2$ and therefore continuous correction of navigation and modelling errors throughout the burn. The requirement to retain both up- and down-throttle margin is why the burn is not flown at full thrust even though full thrust would be propellant-optimal.

6.4 Three-dimensional powered descent guidance

The full problem is to find a thrust history $\mathbf{T}(t)$ that takes the vehicle from state $(\mathbf{r}_0, \mathbf{v}_0, m_0)$ to $(\mathbf{r}_f = \mathbf{0}, \mathbf{v}_f = \mathbf{0})$ at a free final time t_f , using minimum propellant, subject to thrust-magnitude bounds, a glide-slope constraint (stay inside a cone above the pad), and thrust-

pointing constraints. In a flat-Earth landing frame:

$$\min_{t_f, \mathbf{T}(\cdot)} m_0 - m(t_f) \iff \max m(t_f) \quad (30)$$

$$\text{s.t. } \ddot{\mathbf{r}} = \mathbf{g} + \frac{\mathbf{T}}{m}, \quad \dot{m} = -\frac{\|\mathbf{T}\|}{v_e}, \quad (31)$$

$$0 < T_{\min} \leq \|\mathbf{T}(t)\| \leq T_{\max}, \quad (32)$$

$$\mathbf{r}(t) \in \mathcal{C} = \{\mathbf{r} : \|(r_x, r_y)\| \leq r_z \tan \theta_{gs}\}, \quad (33)$$

$$\hat{\mathbf{n}}^\top \mathbf{T}(t) \geq \|\mathbf{T}(t)\| \cos \theta_{pt}, \quad (34)$$

$$\mathbf{r}(t_f) = \mathbf{0}, \mathbf{v}(t_f) = \mathbf{0}, m(t_f) \geq m_{\text{dry}}. \quad (35)$$

Equation (32) is the crux: the lower bound $T_{\min} > 0$ makes the feasible thrust set an *annulus*, which is non-convex. Non-convex problems cannot in general be solved to global optimality in bounded time, which is unacceptable for a flight computer with a few tens of milliseconds per guidance cycle.

6.5 Lossless convexification

The key result, due to Açikmeşe and Ploen (2007) and extended by Açikmeşe, Carson and Blackmore (2013), is that the problem can be made convex without changing its solution. Introduce a slack variable $\Gamma(t)$ and replace (32) by

$$\|\mathbf{T}(t)\| \leq \Gamma(t), \quad T_{\min} \leq \Gamma(t) \leq T_{\max}, \quad \dot{m} = -\Gamma/v_e. \quad (36)$$

The set $\{(\mathbf{T}, \Gamma) : \|\mathbf{T}\| \leq \Gamma\}$ is a second-order cone (convex), and the box on Γ is convex. The relaxed problem is therefore convex. The theorem states:

Theorem 6.1 (Lossless convexification). *Under mild conditions on the dynamics (the pair (A, B) in $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$ is controllable and the cost is monotone in Γ), every optimal solution of the relaxed problem satisfies $\|\mathbf{T}(t)\| = \Gamma(t)$ almost everywhere, and is therefore also an optimal solution of the original non-convex problem.*

Sketch. By Pontryagin’s maximum principle, the Hamiltonian is linear in Γ and in $\mathbf{T}$ with respective costate coefficients. Optimality with respect to $\mathbf{T}$ for fixed Γ requires $\mathbf{T}$ to be aligned with the velocity costate $\boldsymbol{\lambda}_v$ and to have the largest allowed magnitude, i.e. $\|\mathbf{T}\| = \Gamma$, unless $\boldsymbol{\lambda}_v \equiv 0$ on an interval. Controllability rules out $\boldsymbol{\lambda}_v$ vanishing on any interval of positive measure (it would force the entire costate to zero, contradicting the non-triviality condition). Hence the constraint is active almost everywhere. $\square$

The remaining non-convexity, $1/m$ in (31), is removed by the change of variables $z = \ln m$, $\mathbf{u} = \mathbf{T}/m$, $\sigma = \Gamma/m$, giving linear dynamics $\ddot{\mathbf{r}} = \mathbf{g} + \mathbf{u}$, $\dot{z} = -\sigma/v_e$, and the thrust bounds become

$$T_{\min}e^{-z} \leq \sigma \leq T_{\max}e^{-z}. \quad (37)$$

The lower bound $\sigma \geq T_{\min}e^{-z}$ is the epigraph of a convex function and is therefore already a convex constraint; in practice it is replaced by its second-order Taylor expansion about a reference $z_0(t)$ so that it takes second-order-cone form. The upper bound $\sigma \leq T_{\max}e^{-z}$ is *not* convex (a hypograph of a convex function) and is replaced by the first-order expansion $\sigma \leq T_{\max}e^{-z_0}[1 - (z - z_0)]$; because a convex function lies above every tangent line, this approximation is conservative, never permitting more thrust than the engine can deliver. The result is a second-order cone programme (SOCP) with a few thousand variables that interior-point methods solve in milliseconds.

6.6 Flight implementation

Blackmore’s 2016 account describes the Falcon 9 guidance as solving exactly such a convex programme onboard, in real time, using custom code generated by CVXGEN (Mattingley and Boyd), each solve taking well under the guidance-cycle period. The solution is recomputed continuously as the state estimate updates, so that the commanded thrust always corresponds to the propellant-optimal trajectory from the vehicle’s *current* state to the pad. The same convex-optimisation approach was demonstrated on the Masten Xombie test vehicle under NASA’s G-FOLD (Guidance for Fuel-Optimal Large Diverts) programme in 2012–2013 and is now the standard technique for propulsive landing across the industry.

6.7 Landing precision and touchdown

The reported landing accuracy of Falcon 9 is of order 10 m on a 90 m-wide drone-ship deck, from a state at entry-burn cutoff that is uncertain to several hundred metres. The final position error must be smaller than the deck margin *and* the landing legs must arrest the residual velocity. The legs (four, carbon-fibre and aluminium honeycomb) contain crushable aluminium cores that absorb energy irreversibly; for a touchdown at $v_{td} = 2 \text{ m/s}$ the kinetic energy is $\frac{1}{2}(27\,000)(2)^2 \approx 54 \text{ kJ}$, which four crush cores absorb over a stroke of a few centimetres at a mean force of a few hundred kilonewtons. The drone ship adds a further constraint: the deck heaves and rolls in sea state, so the touchdown must occur within the leg’s angular tolerance (a few degrees) of deck-normal; ships hold position within about 3 m using azimuth thrusters and GPS.

The Super Heavy tower catch removes the legs entirely. The booster (mass at catch $\approx 200 \text{ t}$) hovers on three centre Raptors, translates laterally into the closing arms, and settles onto load-bearing pins under its grid fins. The saved leg mass (several tonnes on a vehicle this size) is precisely the kind of gain Eq. (11) rewards exponentially, and the catch was first accomplished on Starship Flight 5 on 13 October 2024.

7 Navigation and Control During Descent

7.1 State estimation

Guidance is only as good as the state estimate it is fed. The booster carries redundant inertial measurement units (IMUs), GPS receivers, and, for the final approach, radar altimeters. These are fused in an extended Kalman filter with state $\mathbf{x} = (\mathbf{r}, \mathbf{v}, \mathbf{q}, \mathbf{b}_a, \mathbf{b}_\omega)$, comprising position, velocity, attitude quaternion, and accelerometer/gyro biases. The prediction step integrates the IMU:

$$\dot{\mathbf{r}} = \mathbf{v}, \quad \dot{\mathbf{v}} = R(\mathbf{q})(\mathbf{a}_{\text{meas}} - \mathbf{b}_a) + \mathbf{g}, \quad \dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes (\boldsymbol{\omega}_{\text{meas}} - \mathbf{b}_\omega), \quad (38)$$

and the update step corrects with GPS position and velocity (1 Hz to 10 Hz) and altimeter range. The critical challenge is the plasma sheath during entry: at $\sim 1.5 \text{ km/s}$ in the lower stratosphere, ionisation is negligible for the booster, so GPS is retained throughout, unlike an orbital re-entry. The IMU-only dead-reckoning error over the 30 s of the landing burn, with a navigation-grade accelerometer bias of $50 \mu\text{g}_0$, is $\frac{1}{2}(50 \times 10^{-6} \times 9.81)(30)^2 \approx 0.22 \text{ m}$, which is why a brief GPS dropout is tolerable near the ground.

7.2 Attitude control authority budget

Three actuator classes are used in sequence:

1. **Cold-gas thrusters** (nitrogen, $I_{sp} \approx 70 \text{ s}$) for the exoatmospheric flip and coast; low thrust, no dependence on dynamic pressure.

2. **Grid fins** for the atmospheric descent; authority scales with $q = \frac{1}{2}\rho v^2$, so they dominate from ~ 60 km down to the landing burn.
3. **Thrust vector control (TVC)** of the gimballed centre engine during all burns; authority is $T\ell_{\text{cg}} \sin \delta_{\text{gim}}$, with ℓ_{cg} the distance from gimbal to centre of mass (only ~ 10 m on an almost-empty booster) and δ_{gim} up to $\sim 5^\circ$.

The hand-over between grid fins and TVC in the final seconds is delicate: as speed drops below ~ 100 m/s, q collapses and the fins lose effectiveness at exactly the moment the vehicle is closest to the ground. The single-engine landing burn provides TVC torque $T\ell_{\text{cg}} \sin 5^\circ \approx 590 \times 10 \times 0.087 \approx 510$ kNm, larger than the grid fins produce at 20 km, which is why control remains crisp to touchdown.

7.3 Linearised pitch dynamics and stability

Near a vertical descent, treating pitch angle θ and lateral position y as small, the linearised dynamics under TVC are

$$I\ddot{\theta} = T\ell_{\text{cg}} \delta - qS\ell_{\text{cp}}C_{N\alpha} \alpha, \quad (39)$$

$$m\ddot{y} = T(\theta + \delta) + qSC_{N\alpha} \alpha, \quad (40)$$

where ℓ_{cp} is the centre-of-pressure offset (destabilising when the centre of pressure is ahead of the centre of mass, which is the case for an engines-first descent with grid fins undeployed, and stabilising with fins deployed). The vehicle is an inverted pendulum balanced on its thrust vector; a PD law $\delta = -k_p\theta - k_d\dot{\theta}$ stabilises it provided the actuator bandwidth (~ 5 Hz for the Merlin gimbal) exceeds the unstable pole $\sqrt{qS\ell_{\text{cp}}C_{N\alpha}/I}$, which for $q = 20$ kPa is of order 0.5 rad/s. The practical design is a gain-scheduled linear-quadratic regulator with q and m as scheduling variables, and structural bending modes (first mode ~ 2 Hz on a 41 m tube) notched out of the loop.

8 Scaling Laws: Why Size Helps

8.1 The square–cube law for tanks

For a cylindrical tank of radius R and length L with wall thickness t set by hoop stress under internal pressure p , $t = pR/\sigma_{\text{allow}}$. The wall mass is $m_w = 2\pi RLt\rho_m = 2\pi R^2Lp\rho_m/\sigma_{\text{allow}}$, while the propellant mass is $m_p = \pi R^2L\rho_p$. Their ratio,

$$\frac{m_w}{m_p} = \frac{2p\rho_m}{\rho_p\sigma_{\text{allow}}}, \quad (41)$$

is *independent* of size for a pressure-stabilised tank. In practice, however, the wall thickness has a manufacturing minimum (~ 2 mm for welded aluminium, ~ 4 mm for stainless steel), and fixed-mass items (avionics, valves, engine controllers, legs, fins, actuators) do not scale with propellant mass. Consequently the structural coefficient ε decreases with vehicle size, roughly as

$$\varepsilon(m_p) \approx \varepsilon_\infty + \frac{m_{\text{fixed}}}{m_p}, \quad (42)$$

with $\varepsilon_\infty \approx 0.03$ – 0.04 and m_{fixed} of order a few tonnes. This is the first reason reusability favours large rockets: the recovery hardware (legs, fins, thrusters, extra thermal protection) is a smaller fraction of a larger stage.

8.2 Ballistic coefficient scaling

For geometrically similar boosters, $m \propto \ell^3$ and $A \propto \ell^2$, so $\beta \propto \ell$: bigger vehicles have higher ballistic coefficients and, by Eq. (25), decelerate at the same peak g but lower in the atmosphere.

Their heating behaves differently. The peak stagnation heat flux $\dot{q}_{s,\max} \propto \sqrt{\rho_{\text{peak}}/r_n} v_e^3$, with $\rho_{\text{peak}} \propto \beta \propto \ell$ and $r_n \propto \ell$, is independent of size; but the integrated heat load per unit area over the entry pulse scales as $\sqrt{\beta} \propto \sqrt{\ell}$, because the heavier vehicle spends its deceleration in denser air. Against this, the thermal capacity per unit area of the wall scales with wall thickness, which for a pressure-designed tank grows as ℓ . The wall temperature rise therefore scales as $\sqrt{\ell}/\ell = \ell^{-1/2}$ and *decreases* with size. Larger boosters can therefore use heavier structural materials (Starship’s 304L stainless steel versus Falcon’s aluminium-lithium) that tolerate higher temperatures, eliminating the entry burn altogether.

8.3 Engine count and landing thrust

A landing booster needs a thrust of order its landing weight, i.e. $T_{\text{land}} \approx (1.2\text{--}2) m_{\text{land}} g_0$. With engines of rated thrust T_e and minimum throttle fraction $\phi_{\min}$, the number of engines that can be lit for landing is

$$n_{\text{land}} \approx \frac{T_{\text{land}}}{\phi_{\min} T_e}, \quad \text{while} \quad n_{\text{launch}} = \frac{n_0 m_0 g_0}{T_e}. \quad (43)$$

The ratio $n_{\text{land}}/n_{\text{launch}} \approx (1.5/\phi_{\min})(m_{\text{land}}/n_0 m_0)$. For Falcon 9, $m_{\text{land}}/m_0 \approx 0.05$, $n_0 = 1.4$, $\phi_{\min} = 0.4$, giving $n_{\text{land}}/n_{\text{launch}} \approx 0.13$, i.e. about one engine of nine. For Super Heavy, $m_{\text{land}}/m_0 \approx 0.04$ and $\phi_{\min} \approx 0.4$ give ~ 3 of 33 engines. Both vehicles do exactly this. The lesson is that a reusable booster wants *many small engines*: a single-engine booster would need a $\sim 20\%$ throttle floor, which is very difficult for a pump-fed engine (combustion instability, injector pressure drop), whereas a nine- or thirty-three-engine cluster achieves the required landing T/W by shutting engines down. Engine-out tolerance on ascent comes for free with the same architecture.

8.4 Small launchers and the Electron problem

Rocket Lab’s Electron (13 t liftoff, 300 kg to LEO) illustrates the other end of the scale. Its first stage has a dry mass of about 950 kg; propulsive landing would require legs, a restart-capable engine with deep throttling (Rutherford’s electric pumps can throttle but the landing thrust of 25 kN on a ~ 1 t stage gives $T/W > 2$ even on one engine), and recovery propellant of ~ 150 kg, all of which would consume essentially the entire payload. Rocket Lab therefore chose passive recovery: an aerodynamically-controlled ballistic entry with a heat shield, decelerating to a drogue and main parachute, followed by a marine splashdown (helicopter capture was attempted in 2022 and abandoned in favour of splashdown). The recovered stage’s ballistic coefficient of only ~ 400 kg/m² means it decelerates high in the atmosphere, where peak heating is low, at the cost of an $\sim 8g$ peak load. Neutron, the company’s 13 t-class successor, is large enough for propulsive landing and is designed around it from the start, with a fixed fairing and an upper stage hung inside the first stage to minimise the recovered structure.

9 Case Studies

9.1 Falcon 9: the reference case

Using the first-principles numbers above we can reconstruct the Falcon 9 Block 5 ASDS mission:

1. **Liftoff**: $m_0 = 549$ t, $T = 7.6$ MN, $n_0 = 1.41$.
2. **MECO** at T+155 s: booster has burned ~ 380 t of 395 t, reserving ~ 15 t (11 t for burns, 4 t for residuals and margin). Δv delivered by the booster, from Eq. (2) with mean $I_{\text{sp}} = 300$ s: $2943 \ln(549/169) \approx 3.47$ km/s; less ~ 1.2 km/s gravity and drag losses gives 2.3 km/s inertial, matching Table 4.

3. **Second stage:** $m_{0,2} = 96 + 17 = 113$ t (stage plus payload), $m_{f,2} = 4 + 17 = 21$ t, $I_{sp} = 348$ s: $\Delta v_2 = 3414 \ln(113/21) = 5.74$ km/s. Adding the booster's 2.3, the 0.41 rotation credit, and subtracting ~ 0.6 km/s of upper-stage gravity loss: 7.85 km/s, essentially orbital velocity at 200 km. The model reproduces the 17 t ASDS payload figure with no free parameters.
4. **Recovery:** entry burn 6.3 t, landing burn 4 t (single engine, 25 s, $\dot{m} \approx 200$ kg/s at 70 %), residuals 1.5 t.

9.2 Falcon Heavy: three simultaneous recoveries

Falcon Heavy's side boosters separate at $\sim T+150$ s at a lower velocity than the centre core (because the centre core throttles down during the side-booster burn to conserve propellant) and return to LZ-1/LZ-2 in a pair of synchronised RTLS trajectories. The centre core continues to a MECO velocity of ~ 3 km/s to 3.5 km/s and attempts an ASDS landing far downrange (900 km to 1000 km). The centre core's higher entry energy ($\mathcal{E} \approx 6$ MJ/kg) pushes it into the regime where entry-burn propellant becomes prohibitive; several centre cores have been expended or lost, and on high-energy missions SpaceX now routinely flies the centre core expendable while recovering the side boosters, a pattern the penalty analysis of Fig. 1 predicts.

9.3 Starship / Super Heavy: full reusability at scale

Super Heavy (~ 3600 t propellant, 9 m diameter, 33 Raptors) uses *hot staging*: Starship's engines ignite before separation, through a vented interstage, so that the booster does not need to coast and flip unpowered. The booster then performs a boostback with ~ 13 engines and re-enters without an entry burn, protected by its steel structure. The landing burn lights 13 engines, then 3, and the final approach is a hover into the tower arms. Starship, the upper stage, is where the new mathematics lies: it re-enters from orbital velocity with a hexagonal-tile heat shield on the windward side, flying a "belly-flop" at $\sim 70^\circ$ angle of attack that maximises drag area ($C_D A \approx 800$ m² for a 50 m $\times$ 9 m body) and lowers the ballistic coefficient to ~ 150 kg/m², followed by a flip to vertical using flap-induced pitch and engine relight in the last kilometre. The full-stack reusable payload target is 100 t to 150 t to LEO, versus ~ 250 t expendable, a 40 % to 60 % penalty that is acceptable only because the marginal cost per flight is intended to be propellant plus refurbishment.

9.4 New Shepard and New Glenn: hover-capable landing

New Shepard is a single-stage suborbital vehicle: 75 t at liftoff, BE-3 hydrogen engine, apogee ~ 105 km, peak speed ~ 1 km/s. Its recovery is the trivial case of the analysis above: entry energy 1.5 MJ/kg, no boostback or entry burn (it lands where it launched, because it goes straight up), deep throttling to 20 % so that $T/W < 1$ is available, allowing a slow, hovering approach with ring-fin and wedge-fin aerodynamic stabilisation. New Glenn (7 m diameter, seven BE-4 methane engines, ~ 17 MN) is an orbital-class two-stage vehicle whose first stage flies a Falcon-like ASDS profile at somewhat higher staging velocity, with aerodynamic strakes providing body lift and fins at the aft end for control. Its booster was recovered on the drone ship *Jacklyn* on the vehicle's second flight in November 2025, the first orbital-class booster landing by a company other than SpaceX.

10 Multi-Flight Life: Structures, Thermal Cycling, and Engines

10.1 Fatigue and the flight-count limit

Each flight subjects the tank walls to a pressurisation cycle ($p \approx 3$ bar over ambient on Falcon 9), a thermal cycle (-183°C to 150°C for LOX tanks), and dynamic loads from max- Q and the

entry pulse. For aluminium-lithium alloys in the low-cycle regime, the Basquin–Coffin–Manson relation

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma'_f}{E}(2N_f)^b + \varepsilon'_f(2N_f)^c \quad (44)$$

with typical constants ($b \approx -0.1$, $c \approx -0.6$) predicts $N_f > 10^3$ cycles at the strain ranges seen in tanks designed with a factor of safety of 1.25–1.4 on limit load, so fatigue of the primary structure is not what limits the flight count. SpaceX initially certified Block 5 for 10 flights, then extended it to 15, 20, and by 2025–2026 to 40 flights, on the basis of accumulated inspection data rather than a change in design.

10.2 Engine life

The Merlin 1D is the pacing item. Its gas-generator turbopump runs at $\sim 36\,000$ rpm with turbine inlet temperatures near 800°C ; each start is a thermal shock and each shutdown a coking event. Kerosene coking is why methane was chosen for Raptor and BE-4. A booster flight involves four burns (ascent, boostback, entry, landing) on the centre engine and two on the outer engines, so a 30-flight booster’s centre engine has seen ~ 120 starts. Engines are hot-fired between flights, inspected borescopically, and swapped as needed; the effective “engine flight count” is decoupled from the airframe flight count by this rotation.

10.3 Landing loads and the 2 m/s touchdown

The largest single-event load on the recovered structure is touchdown. With $m_{\text{land}} = 27\text{ t}$, $v_{\text{td}} \leq 2\text{ m/s}$, and a crush stroke $s = 0.1\text{ m}$, the mean deceleration is $v^2/(2s) = 20\text{ m/s}^2$ and the peak leg load is $\sim 2g \times$ landing weight $\approx 540\text{ kN}$ shared among four legs. The crush cores are consumable, replaced between flights; everything else is designed to survive this load elastically for the life of the vehicle. A tower catch shifts this load to the booster’s forward structure through the catch pins under the grid fins, which are reinforced for a $\sim 3g$ dynamic factor on the 200 t catch mass, i.e. $\sim 6\text{ MN}$ total.

11 Economics: When Reuse Pays

11.1 Cost per flight model

Let C_B be the cost of manufacturing a booster, C_U the cost of the (expended) upper stage and fairing, C_R the refurbishment and recovery cost per flight (drone ship operations, inspection, crush cores, engine swaps), N the number of flights per booster, and π the payload penalty fraction. The cost per kilogram to orbit is

$$\frac{C}{m_L} = \frac{C_B/N + C_R + C_U}{(1 - \pi) m_{L,\text{exp}}}. \quad (45)$$

Reuse is favourable when $C_B/N + C_R < C_B - \pi(C_B + C_U)/(1 - \pi)$, i.e. roughly when

$$N > \frac{C_B}{C_B(1 - \pi) - \pi C_U - C_R(1 - \pi)}. \quad (46)$$

Example 11.1 (Falcon 9 breakeven). *Public estimates put the Falcon 9 booster at $\sim 60\%$ of a $\sim \$60\text{M}$ vehicle cost ($C_B \approx \$36\text{M}$), upper stage plus fairing at $\sim \$15\text{M}$ (C_U ; fairings are now also recovered), and C_R at $\$1\text{--}3\text{M}$. With $\pi = 0.28$: $N > 36/(36 \times 0.72 - 0.28 \times 15 - 2 \times 0.72) \approx 36/20.3 \approx 1.8$. Breakeven occurs on the second flight, and at $N = 20$ the booster’s amortised cost per flight is under $\$2\text{M}$, about 3% of the vehicle’s expendable cost. The marginal cost of a Falcon 9 launch has been reported by SpaceX at roughly $\$15\text{--}28\text{M}$, dominated by the expended upper stage, which is why Starship’s design brief was full reusability.*

11.2 The upper-stage bottleneck

Equation (45) shows that as $N \rightarrow \infty$, cost per kilogram asymptotes to $(C_R + C_U)/((1 - \pi)m_{L,\text{exp}})$. With C_U a substantial fraction of the total, booster reuse alone cannot reduce cost per kilogram below roughly one-third of the expendable value. Reducing C_U requires either a very cheap expendable upper stage (the Falcon approach: a single Merlin Vacuum, common tooling with the booster) or a reusable one (the Starship approach). The penalty π for reusing an upper stage is large, as discussed in Section 3, which is why Starship pays it with sheer scale.

11.3 Cadence as the hidden variable

The cost model omits fixed costs (pads, ranges, staff), which are large: a launch provider with annual fixed costs F and n launches per year pays F/n per launch. Reuse raises n for a fixed manufacturing rate: SpaceX built roughly ten new boosters per year while flying over 130 Falcon missions in 2024 and 160+ in 2025. High cadence in turn generates the inspection and telemetry data that justify extending N , in a positive feedback loop that expendable vehicles cannot enter.

12 Conclusion

The physics of reusable rockets is the physics of ordinary rockets, seen through the exponential lens of the Tsiolkovsky equation. Reserving 4 % of a booster’s propellant for recovery costs 25 % to 40 % of the payload because that reserve is amplified through the upper stage; the same exponential makes the recovered hardware worth 97 % of its manufacturing cost after twenty flights. Between MECO and touchdown, the booster’s kinetic energy is removed mainly by the atmosphere in accordance with the Allen–Eggers ballistic entry solution, with an entry burn used to hold the peak dynamic pressure and heating within what an ascent-optimised structure can tolerate. The final landing is a genuine real-time optimal-control problem, made solvable by lossless convexification, executed on the flight computer as a second-order cone programme, and actuated by a gimballed engine that balances the vehicle like an inverted pendulum. The scaling laws favour large vehicles with many small engines, deep throttling, and heavy-material structures; the economics favour any vehicle that can achieve high cadence.

Every number in this paper for Falcon 9 was obtained from published mass, thrust, and I_{sp} values through elementary physics, and reproduces the observed staging velocities, burn durations, and payload capabilities to within a few percent. What remained hard for sixty years was not the mathematics but its integration: throttleable, restartable engines; navigation that survives the flip and entry; control authority through the transonic regime; and, above all, the organisational willingness to fly, fail, and iterate. That integration is now the industry baseline, and the next problems (upper-stage reuse, rapid turnaround, and orbital refuelling) are again problems of mass ratio, heat, and control.

Acknowledgements and Data Note

Vehicle parameters are compiled from manufacturer specification pages, launch webcasts, FAA environmental assessments, and conference publications; they are approximate and change with vehicle block and mission. All derivations are the author’s own and any errors are the author’s responsibility. Correspondence: kye@swarms.world.

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